(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :19/08/2024

(21) Application No.202441062415 A

(43) Publication Date : 30/08/2024

(54) Title of the invention : A SYSTEM FOR POWERING AND LEVERAGING ENERGY HARVESTING TECHNIQUES FOR SUSTAINABLE WIRELESS SENSOR NETWORKS IN LOW POWER WAN DEPLOYMENTS THROUGH EDGE

(51) International classification	:H04W0084180000, H04W0004380000, H04W0004700000, H04L0067120000, H02J0007340000
(86) International Application No	:NA
Filing Date	:NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number	:NA
Filing Date	:NA
(62) Divisional to Application Number	:NA
Filing Date	:NA

(71)**Name of Applicant :**

1)S.LOGANAATH

Address of Applicant :B.E.ECE-II YEAR, VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY, MADURAI, TAMIL NADU, INDIA - 625009. ---------- ----------

2)R.KARTHI KUMAR

3)M.KARTHIKEYA

4)T.J.UMESHBALAJI

5)J.THYA

6)B.SUVEDHA

7)S.LAKSHIKA

8)T.R.ARUN

9)R.P.NIRANJAN KUMAR

10)R.P.NIRMAL KUMAR

Name of Applicant : NA

Address of Applicant : NA

(72)**Name of Inventor :**

1)S.LOGANAATH

Address of Applicant :B.E.ECE-II YEAR, VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY, MADURAI, TAMIL NADU, INDIA - 625009. ---------- ----------

2)R.KARTHI KUMAR

Address of Applicant :ASSISTANT PROFESSOR, DEPARTMENT OF ECE, VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY, MADURAI, TAMIL NADU , INDIA-625009. ---------- ----------

3)M.KARTHIKEYA

Address of Applicant :B.E-ECE-II YEAR, VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY, MADURAI, TAMIL NADU , INDIA-625009. ---------- ----------

4)T.J.UMESHBALAJI

Address of Applicant :B.E-ECE-II YEAR, VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY, MADURAI, TAMIL NADU , INDIA-625009. ---------- ----------

5)J.THYA

Address of Applicant :B.E-ECE-II YEAR, VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY, MADURAI, TAMIL NADU , INDIA-625009. ---------- ----------

6)B.SUVEDHA

Address of Applicant :B.E-ECE-II YEAR, VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY, MADURAI, TAMIL NADU , INDIA-625009 ---------- ----------

7)S.LAKSHIKA

Address of Applicant :B.E-ECE-II YEAR, VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY, MADURAI, TAMIL NADU , INDIA-625009. ---------- ----------

8)T.R.ARUN

Address of Applicant :B.E-ECE-II YEAR, VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY, MADURAI, TAMIL NADU , INDIA-625009. ---------- ----------

9)R.P.NIRANJAN KUMAR

Address of Applicant :B.E-ECE-II YEAR, VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY, MADURAI, TAMIL NADU , INDIA-625009. ---------- ----------

10)R.P.NIRMAL KUMAR

Address of Applicant :B.E-ECE-II YEAR, VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY, MADURAI, TAMIL NADU , INDIA-625009. ---------- ----------

(57) Abstract :

The rise of the Internet of Things (IoT) has ushered in an era of pervasive connectivity and data collection. At the forefront of this transformation are Wireless Sensor Networks (WSNs), essential for monitoring and controlling various environments and processes. However, the sustainability of WSNs, especially in Low Power Wide Area Network (LPWAN) deployments, is significantly influenced by efficient energy management. Energy harvesting techniques offer a promising solution, enabling continuous sensor node operation with minimal reliance on conventional power sources. This invention examines the technical nuances and challenges of leveraging energy harvesting techniques to power edge devices in LPWAN deployments and provides a system to address the said problem. It discloses WSNs' structure, LPWAN technologies like LoRaWAN, and the challenges of managing-limited energy resources in sensor nodes. The invention then provides solar, thermal, kinetic, and RF energy harvesting techniques, emphasizing their advantages and limitations. Finally, the invention highlights the integration of these methods into a cohesive hybrid energy harvesting system to ensure reliable and sustainable WSN operations. By integrating multiple energy harvesting modules, WSNs can achieve perpetual operation, reducing maintenance costs and environmental impact. The invention also provides the optimization of energy harvesting in LPWAN deployments, including the selection of optimal energy harvesting modules based on environmental conditions. Overall, energy harvesting has the potential to revolutionize the sustainability of WSNs in LoRaWAN deployments, enabling widespread adoption of IoT applications. FIG 1

No. of Pages : 19 No. of Claims : 5

Scanned with OKEN Scanner